

I-V characteristics → Problem 13

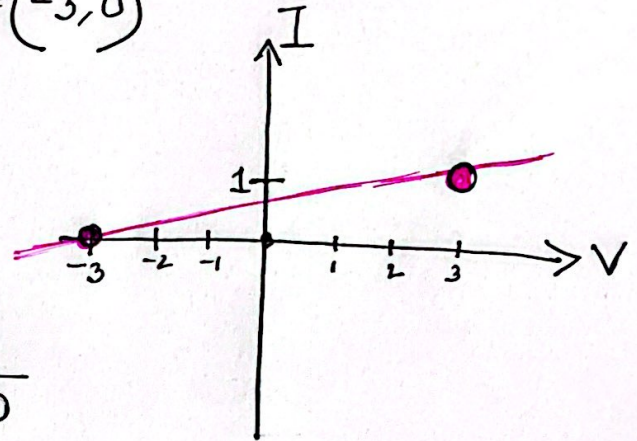
data points

① when applied " $V = 3$ " then current " $I = 1$ " ⇒ $(3, 1)$
and, " " $V = -3$ " then current " $I = 0$ " ⇒ $(-3, 0)$

$$\therefore (V_1, I_1) = (3, 1) \text{ and } (V_2, I_2) = (-3, 0)$$

② and, ③
I-V equation derive -

$$\frac{V - V_1}{V_1 - V_2} = \frac{I - I_1}{I_1 - I_2} \Rightarrow \frac{V - 3}{3 - (-3)} = \frac{I - 1}{1 - 0}$$



$$\Rightarrow \frac{V - 3}{6} = \frac{I - 1}{1}$$

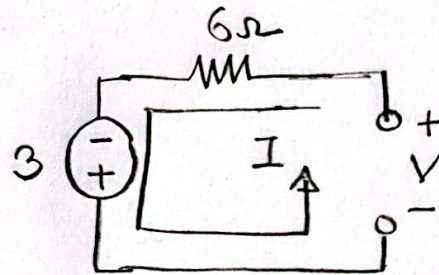
drawing the KVL equation →

$$\Rightarrow V - 3 = 6I - 6$$

$$\Rightarrow 0 = -V + 6I - 3$$

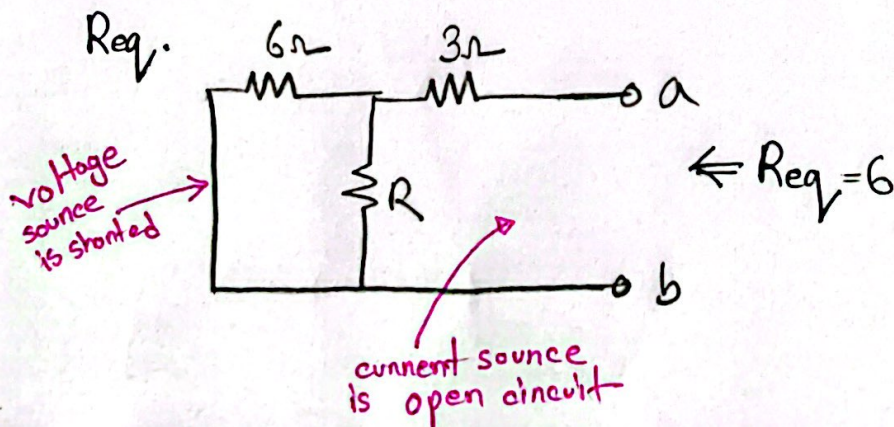
$$\therefore \boxed{-V + 6I - 3 = 0}$$

is a KVL equation



thus, we can see, $R_{eq} = 6\Omega$

so, to find ' R ' from the bigger (un-simplified) circuit, we consider that all of its resistances are combined into



$$\therefore 3 + (6 \parallel R) = R_{eq}$$

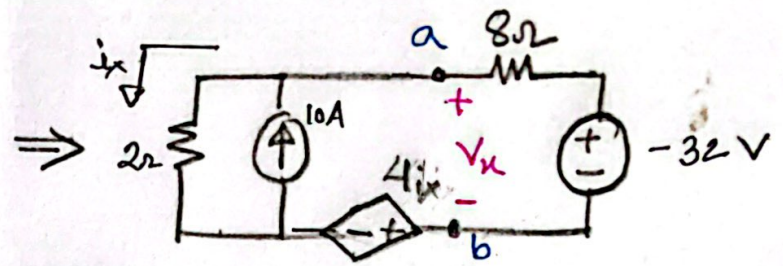
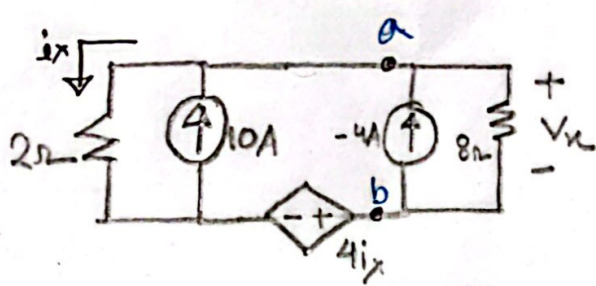
$$\Rightarrow 6 \parallel R = 6 - 3$$

$$\Rightarrow \frac{6R}{6 + R} = 3$$

$$\Rightarrow 6R = 18 + 3R$$

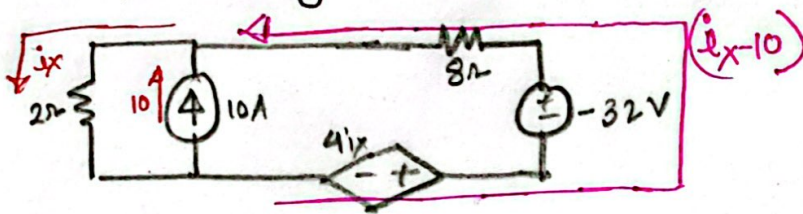
$$\Rightarrow 3R = 18 \Rightarrow R = 6$$

Source Transformation $\rightarrow$ Problem 10

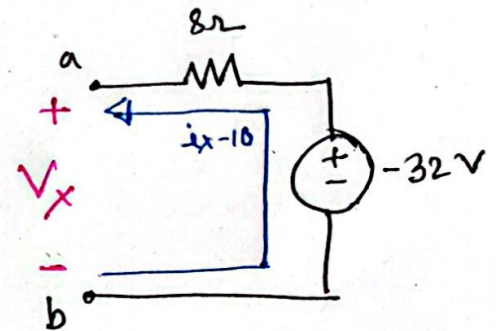


even with source transforming, the ' V_x ' is still remains in the same nodes, that's why we carry on with source transformation

Now, let's figure out the current,



thus



Now, KVL at circuit $\Rightarrow$

$$2i_x - 4i_x - V_n = 0$$

$$\Rightarrow -V_n = 2i_x$$

$$\Rightarrow \boxed{i_x = \frac{V_n}{-2}} \dots \textcircled{\text{ii}}$$

$$\begin{aligned} \therefore V_x &= (-V_{8\Omega}) + (-32) \\ &= -\{8 \times (i_x - 10)\} - 32 \\ &= -8i_x + 80 - 32 \end{aligned}$$

$$\boxed{V_n = 48 - 8i_x} \dots \textcircled{\text{i}}$$

from, $\textcircled{\text{i}}$ and $\textcircled{\text{ii}}$ we get, $V_n = 48 - 8\left(\frac{V_n}{-2}\right)$

$$\Rightarrow V_n = 48 + 4V_n$$

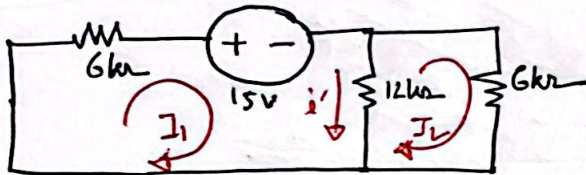
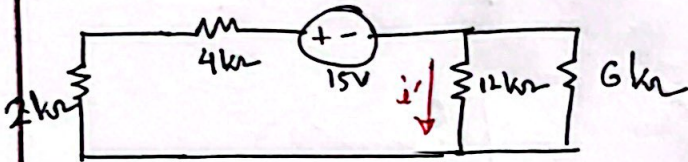
$$\Rightarrow V_n - 4V_n = 48$$

$$V_n = \frac{48}{-3}$$

$$\boxed{V_n = -16V}$$

Superposition --> Problem 3

Ⓐ Superposition for 15V



Mesh 1

$$6I_1 + 15 + 12(I_1 - I_2) = 0$$

$$\Rightarrow 18I_1 - 12I_2 = -15$$

$$\Rightarrow \boxed{6I_1 - 4I_2 = -5}$$

Mesh 2

$$6I_2 + 12(I_2 - I_1) = 0$$

$$\Rightarrow -12I_1 + 18I_2 = 0$$

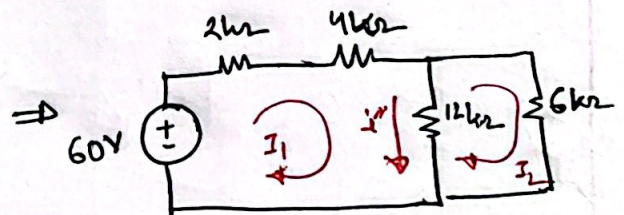
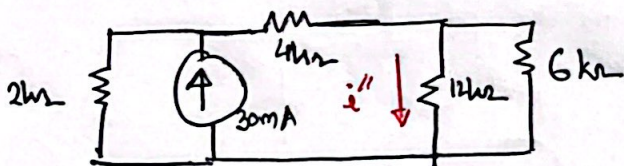
$$\Rightarrow \boxed{-2I_1 + 3I_2 = 0}$$

$$\therefore I_1 = -1.5 \text{ mA}$$

$$I_2 = -1 \text{ mA}$$

$$i' = I_1 - I_2 = \frac{-1.5 - (-1)}{1} = \boxed{-0.5 \text{ mA}}$$

Ⓑ Superposition for 30mA



Mesh 1

$$-60 + 2I_1 + 4I_1 + 12(I_1 - I_2) = 0$$

$$\Rightarrow 18I_1 - 12I_2 = 60$$

$$\Rightarrow \boxed{3I_1 - 2I_2 = 10}$$

Mesh 2

$$6I_2 + 12(I_2 - I_1) = 0$$

$$-12I_1 + 18I_2 = 0$$

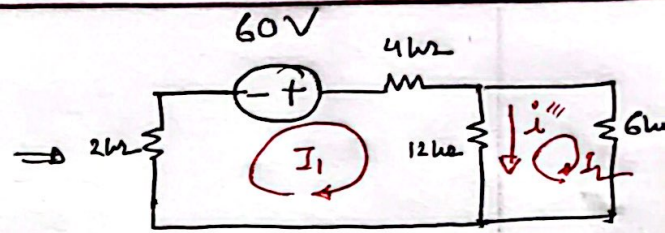
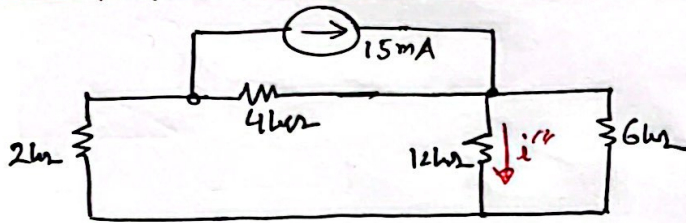
$$\Rightarrow \boxed{-2I_1 + 3I_2 = 0}$$

$$I_1 = 6 \text{ mA}$$

$$I_2 = 4 \text{ mA}$$

$$\therefore i'' = I_1 - I_2 = 6 - 4 = \boxed{2 \text{ mA}}$$

④ Superposition for 15 mA



Mesh 1

$$2I_1 - 60 + 4I_1 + 12(I_1 - I_2) = 0$$

$$\Rightarrow 18I_1 - 12I_2 = 60$$

$$\Rightarrow \boxed{3I_1 - 2I_2 = 10}$$

$$I_1 = 6 \text{ mA}$$

$$I_2 = 4 \text{ mA}$$

Mesh 2

$$6I_2 + 12(I_2 - I_1) = 0$$

$$-12I_1 + 18I_2 = 0$$

$$\Rightarrow \boxed{-2I_1 + 3I_2 = 0}$$

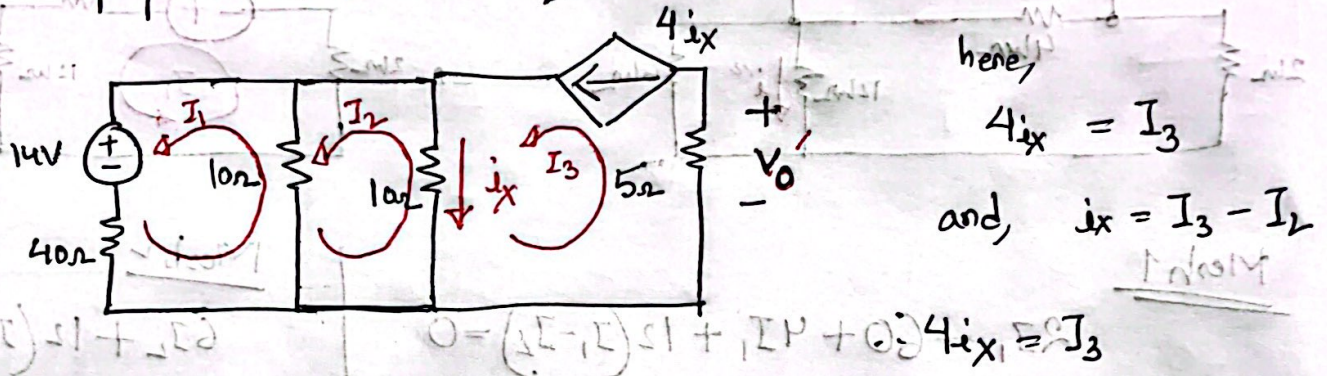
$$\therefore i_o''' = I_1 - I_2 = 6 - 4 = \boxed{2 \text{ mA}}$$

$$\therefore i_o = i' + i'' + i''' = -0.5 + 2 + 2 = \boxed{3.5 \text{ mA}}$$

Problem 7

Superposition --> Problem 7

Superposition for 14V $\Rightarrow$



at mesh 1

$$14 + 40I_1 + 10(I_1 - I_2) = 0$$

$$\Rightarrow 50I_1 - 10I_2 = -14 \quad \dots \textcircled{ii}$$

$$\Rightarrow 4(I_3 - I_2) = I_3$$

$$\Rightarrow -4I_2 + 3I_3 = 0 \quad \dots \textcircled{i}$$

at mesh 2

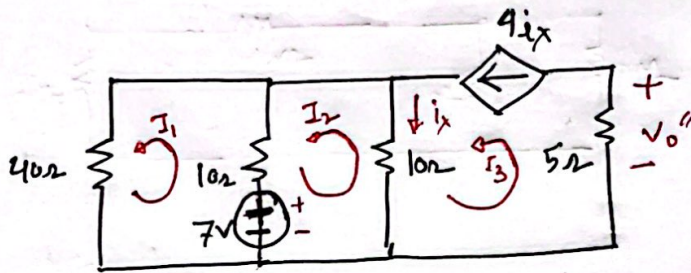
$$10(I_2 - I_1) + 10(I_2 - I_3) = 0$$

$$\Rightarrow -10I_1 + 20I_2 - 10I_3 = 0 \quad \dots \textcircled{iii}$$

$$I_1 = -0.4 \text{ A} ; \quad I_2 = -0.6 \text{ A} ; \quad I_3 = -0.8 \text{ A}$$

$$\therefore V_0' = 5 \times (-I_3) = 5 \times (-(-0.8)) = 5 \times (+0.8) = 4 \text{ V}$$

superposition for 7V $\Rightarrow$



here, $4i_x = I_3$

and, $i_x = I_3 - I_2$

$\Rightarrow 4(I_3 - I_2) = I_3$

$\therefore -4I_2 + 3I_3 = 0$

at mesh 1

$40I_1 - 7 + 10(I_1 - I_2) = 0$

$\Rightarrow 50I_1 - 10I_2 = 7 \dots (ii)$

at mesh 2

$10(I_2 - I_1) + 7 + 10(I_2 - I_3) = 0$

$\Rightarrow -10I_1 + 20I_2 - 10I_3 = -7 \dots (iii)$

$I_1 = 0.5A ; I_2 = 1.8A ; I_3 = 2.4A$

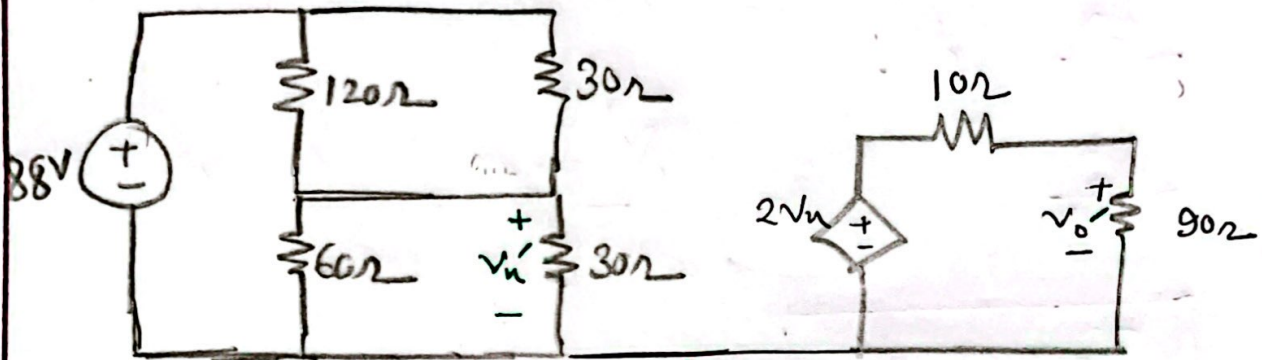
$\therefore V_o'' = 5 \times (-I_3) = 5 \times (-2.4)$
 $= -12V$

$\therefore V_o = V_o' + V_o''$

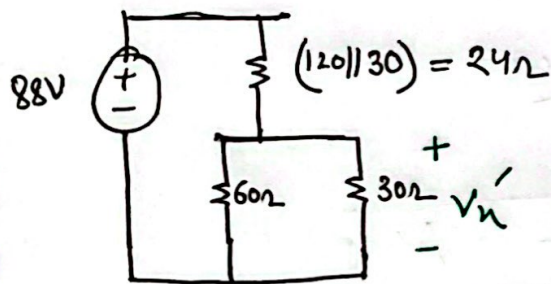
$= 4 + (-12) = -8V$

Superposition for 88V $\Rightarrow$

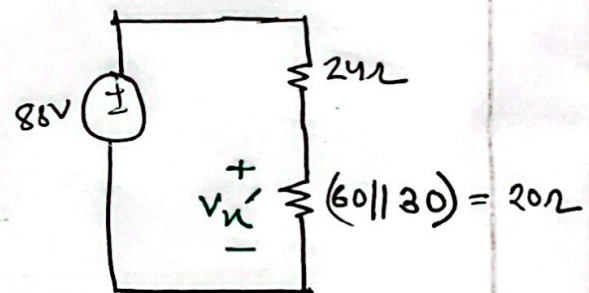
Superposition $\rightarrow$ Problem 8



no current will flow in this line/wire
(so both circuits are separate)



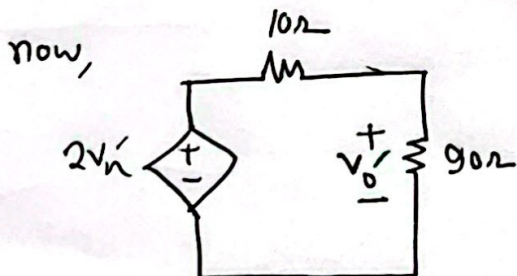
$\Rightarrow$



VDR

$$\therefore v_n' = \frac{20}{20+24} \times 88$$

$$= 40 \text{ V}$$



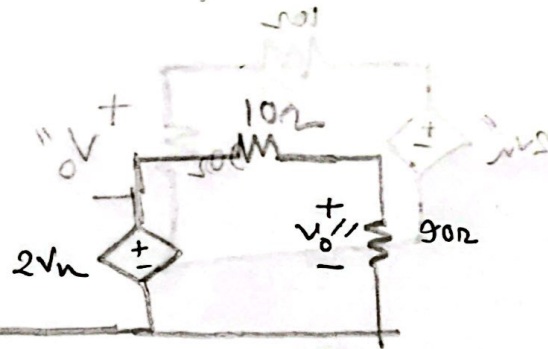
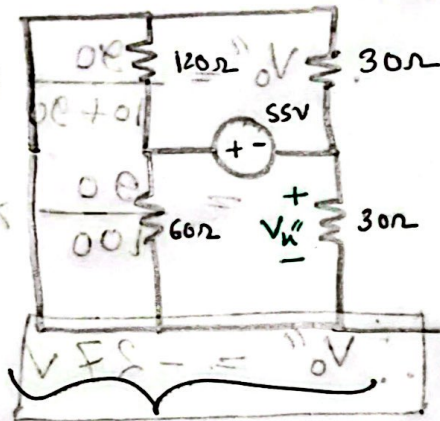
VDR

$$v_o' = \frac{90}{10+90} \times 2 \cdot v_n'$$

$$v_o' = \frac{90}{100} \times 2 \cdot 40$$

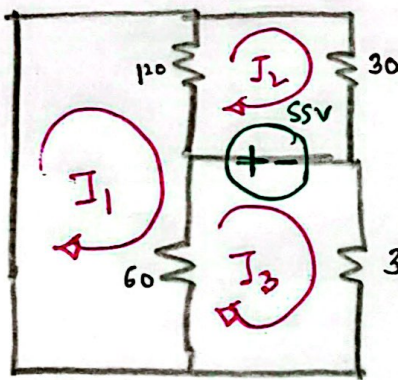
$$v_o' = 72 \text{ V}$$

Superposition for 55V $\Rightarrow$



no current will flow in this line/wire
(so both circuits are separate)

"0V + 0V = 0V total"



Mesh 1
 $120(I_1 - I_2) + 60(I_1 - I_3) = 0$
 $\Rightarrow 180I_1 - 120I_2 - 60I_3 = 0$

$3I_1 - 2I_2 - I_3 = 0$ (i)

Mesh 2
 $30I_2 - 55 + 120(I_2 - I_1) = 0$

$\Rightarrow 150I_2 - 55 - 120I_1 = 0$

$\Rightarrow -120I_1 + 150I_2 = 55$ (ii)

Mesh 3

$60(I_3 - I_1) + 55 + 30I_3 = 0$

$\Rightarrow -60I_1 + 90I_3 = -55$ (iii)

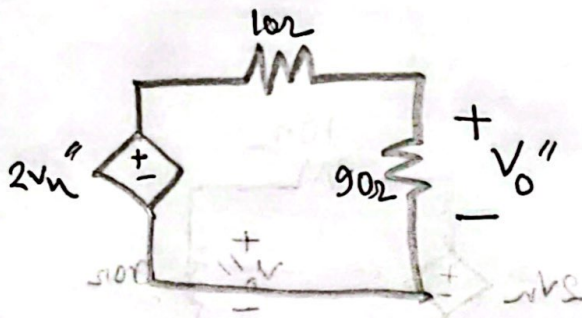
Solving

$\begin{cases} I_1 = 0.1667 A \\ I_2 = 0.5 A \\ I_3 = -0.5 A \end{cases}$

$\therefore V_n = (30\Omega) \cdot I_3$
 $= 30 \times (-0.5)$
 $= -15V$

superposition for VDR

now,



$$\therefore V_0'' = \frac{90}{10+90} \times 2 \cdot V_n''$$

$$= \frac{90}{100} \times 2 \times (-15)$$

$$\therefore V_0'' = -27V$$

$$\therefore \text{total } V_0 = V_0' + V_0''$$

$$= (72 - 27)$$

$$V_0 = -27 + 72$$

Mesh

$$0 = (I_1 - I_2)10 + 22 - 150(I_1 - I_2) = 0$$

$$0 = 120I_1 - 22 - 150I_2 = 0$$

$$-150I_1 + 120I_2 = 22$$

$$0 = 90(I_3 - I_2) + 22 + 30I_3 = 0$$

$$-90I_1 + 90I_2 + 30I_3 = -22$$

Mesh

$$\begin{cases} I_1 = 0.162A \\ I_2 = 0.2A \\ I_3 = -0.2A \end{cases}$$

$$V_n'' = (30I_2) \cdot 13$$

$$= 30 \times (-0.2)$$

AC fundamentals $\Rightarrow$ Problem 13

given, $\Delta t = 30 \text{ ms}$ (considering both curves as sin curves)

$$\text{thus, } \frac{\varphi}{\omega} = \Delta t$$

$$\Rightarrow \varphi = \Delta t \cdot \omega$$

$$\therefore \varphi = (30 \times 10^{-3}) \cdot 25\pi$$

$$\varphi = \underbrace{\frac{3}{4}\pi}_{\text{radians}} \text{ or } \underbrace{135^\circ}_{\text{degrees}}$$

$$\text{here, } T = 80 \text{ ms}$$

$$\therefore \omega = 2\pi f = 2\pi \frac{1}{T}$$

$$= 2\pi \cdot \frac{1}{80 \times 10^{-3}}$$

$$= 25\pi$$

$\therefore V_1(t)$ leads $V_2(t)$ by 135° degrees

as, $V_1(t)$ curve starts from origin, and amplitude '4'

$$\text{thus, } V_1(t) = 4 \sin(25\pi t \pm 0^\circ)$$

$$\boxed{V_1(t) = 4 \sin(25\pi t) \text{ V}}$$

$$\boxed{A_m \cdot \sin(\omega t \pm \varphi)}$$

and, as $V_2(t)$ curve starts 135° right shifted to origin,
and amplitude '2', thus,

$$\boxed{V_2(t) = 2 \sin(25\pi t - 135^\circ) \text{ V}}$$

Problem 8

$$v(t) = 80 \cos(1000\pi t - 30^\circ) \text{ V}$$

a let's assume at $t = t_1$ sec time, the voltage value is 80 V

$$\therefore v(t_1) = 80 \cos(1000\pi t_1 - 30^\circ) = 80$$

$$\Rightarrow \cos(1000\pi t_1 - 30^\circ) = \frac{80}{80}$$

$$\Rightarrow 1000\pi t_1 - 30^\circ = \cos^{-1}\left(\frac{80}{80}\right)$$

$$\Rightarrow 1000\pi t_1 - \frac{\pi}{6} = \cos^{-1}(1)$$

$$\Rightarrow 1000\pi t_1 = 0 + \frac{\pi}{6}$$

$$\Rightarrow t_1 = \frac{\frac{\pi}{6}}{1000\pi}$$

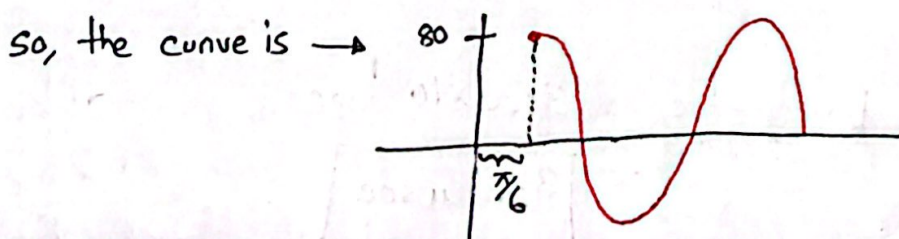
$$\therefore t_1 = 1.667 \times 10^{-4} \text{ sec}$$

$$\boxed{t_1 = 166.7 \text{ } \mu\text{sec}}$$

b the expression is, $v(t) = 80 \cos(1000\pi t - 30^\circ)$

we compare this with, $v(t) = A_m \cdot \cos(\omega t - \varphi)$

$$\text{then, } A_m = 80 ; \quad \omega = 1000\pi ; \quad \varphi = 30^\circ = \frac{\pi}{6}$$



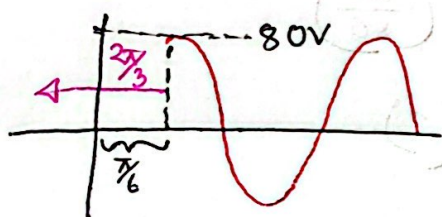
thus, $\frac{2}{3}$ ms left shift means, $\Delta t = \frac{2}{3} \times 10^{-3}$ sec

$$\therefore \varphi = \omega \cdot \Delta t$$

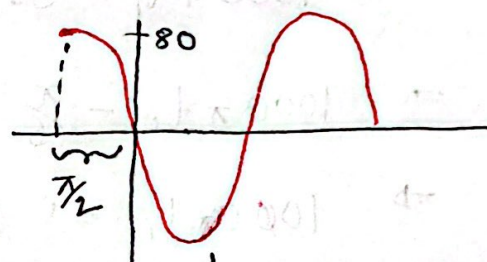
$$= (1000 \pi) \left(\frac{2}{3} \times 10^{-3} \right)$$

$$\varphi = \frac{2\pi}{3}$$

so, we need to left shift the curve $\frac{2\pi}{3}$ from its previous position,



$\Rightarrow$



$$\left(\frac{2\pi}{3} - \frac{\pi}{6} \right) = \frac{4\pi - \pi}{6}$$

$$= \frac{\pi}{2}$$

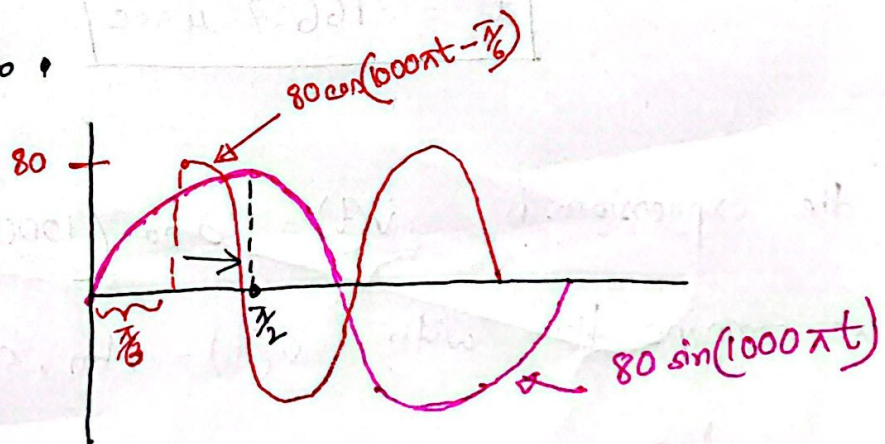
thus, new expression,

$$v(t) = 80 \cos(1000\pi t + 90^\circ)$$

$$= -80 \sin(1000\pi t)$$

C 'cos curve needs to

right shift $\left(\frac{\pi}{2} - \frac{\pi}{6} \right)$
 $= \frac{\pi}{3}$ to turn into
 'sin'

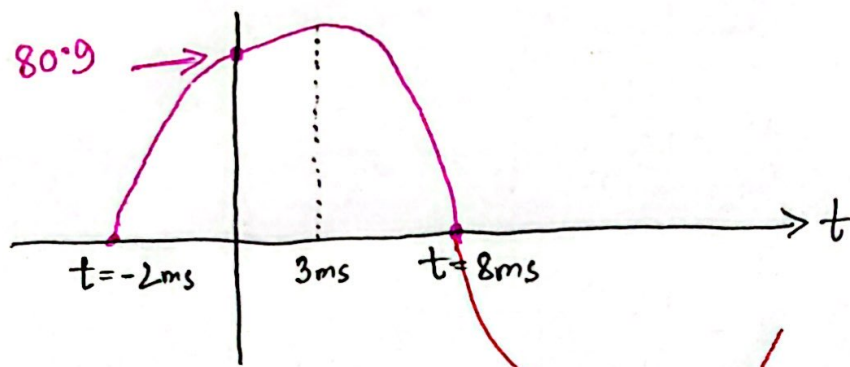


$$\therefore \frac{\pi/3}{\omega} = t$$

$$\Rightarrow \frac{\pi/3}{1000\pi} = t \quad \Rightarrow \quad t = \frac{1}{3 \times 1000} = 3.33 \times 10^{-4} \text{ sec}$$

$$= \boxed{333 \text{ } \mu\text{sec}}$$

Problem 9



here positive half cycle

$$\text{duration} = 8 - (-2) = 10 \text{ ms}$$

$$\therefore \text{Full cycle duration} = T = 20 \text{ ms}$$

$$\therefore \omega = 2\pi f = 2\pi \cdot \frac{1}{T} = \frac{2\pi}{20 \times 10^{-3}} = 100\pi \text{ rad/sec}$$

So, the expression stands $\rightarrow$ (assuming 'cos' wave)

$$A_m \cos(\omega t - \varphi) \\ \Rightarrow A_m \cos(100\pi t - \varphi)$$

$$\frac{\varphi}{\omega} = 3 \text{ ms}$$

$$\Rightarrow \frac{\varphi}{100\pi} = 3 \times 10^{-3}$$

$$\Rightarrow \varphi = (3 \times 10^{-3}) \cdot 100\pi$$

$$\varphi = \frac{3\pi}{10}$$

$$\varphi = 54^\circ$$

and,

$$v(t=0) = 80.9 = A_m \cos(100\pi \cdot 0 - 54^\circ)$$

$$\Rightarrow 80.9 = A_m \cdot \cos(-54^\circ)$$

$$\Rightarrow A_m = \frac{80.9}{\cos(-54^\circ)}$$

$$A_m = 137.63$$

$$\therefore A_m \cos(\omega t - \varphi) = 137 \cos(100\pi t - 54^\circ)$$